

Practice Quiz: Seeing What Others Miss

Part A: Multiple Choice

1. In Active Inference, perception is primarily: A) A bottom-up process where sensory data flows to the brain for interpretation B) A top-down process where the brain generates predictions and processes the difference between predictions and observations C) A random process where the brain samples stimuli without expectations D) A passive process that requires no prior knowledge
2. Why did the rolling suitcase take so long to be invented despite both wheels and suitcases existing for centuries? A) The technology for small wheels did not exist B) Airlines prohibited wheeled luggage until 1970 C) Travelers' generative models had accommodated the difficulty of carrying luggage as normal, suppressing the prediction error D) Patent laws prevented anyone from combining wheels and suitcases
3. Takuo Aoyagi's invention of the pulse oximeter demonstrates which perceptual principle? A) More data always leads to better inventions B) The distinction between signal and noise depends on the observer's generative model C) All fluctuations in measurement are noise that should be filtered out D) Medical inventions can only be made by doctors
4. An inventor who assigns too much weight to their own personal frustration when identifying problems is making which precision weighting error? A) Overweighting quantitative data B) Underweighting expert opinion C) Overweighting personal experience without checking whether the problem generalizes D) Underweighting personal experience
5. IDEO's approach to redesigning the shopping cart — sending teams to observe shoppers before brainstorming — is an example of: A) Passive data collection B) Active sensing, where observations are chosen to maximally reduce uncertainty about real user problems C) Market research through surveys D) Competitive analysis of existing cart manufacturers

6. A chronic prediction error is: A) A problem that occurs only once and never repeats B) A repeated problem that people have learned to accommodate rather than solve, making it invisible to them C) An error in a scientific prediction that has been published D) A manufacturing defect that increases over time
7. Which of the following is the BEST evidence that a perceived problem is genuine signal rather than noise? A) The inventor personally finds it frustrating B) The problem is persistent, experienced by multiple people, caused by design rather than behavior, and potentially solvable C) A focus group agrees the problem exists D) The problem has never been written about before

Part B: Short Answer and Analysis

1. A software developer notices that their elderly parent struggles to use the TV remote — pressing the wrong buttons, accidentally changing inputs, and getting frustrated. The developer assumes this is a widespread problem and begins designing a simplified universal remote. Using the Active Inference framework of evidence gathering as active sensing, propose three specific observations or data-gathering activities the developer should conduct before committing to the invention. For each, explain what information it would provide and how it would update the developer's beliefs about whether this problem is signal or noise.
2. Explain the difference between "fresh eyes" perception and "expert" perception. Sara Blakely (outsider to the hosiery industry) and James Dyson (outsider to the vacuum industry) both benefited from fresh eyes. But many inventions also come from deep experts who notice subtle anomalies that outsiders would miss (like Aoyagi's pulse oximeter). When should an inventor seek fresh eyes, and when should they rely on domain expertise? Is there a way to cultivate both?
3. You are designing a new public park bench. Describe three different "empathic perception" exercises you could conduct — each adopting the generative model of a different user type — to identify prediction errors that your own model would miss. For each exercise, identify the user type, the likely prediction errors they would experience with existing benches, and how this perception would change your design.